

AN AI POWERED SOFTWARE-BASED FIRE DETECTION SYSTEM FOR CCTV PLATFORMS.

FiremeX - TRACEABILITY MATRIX

SYSTEM FUNCTIONAL TESTING

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1. Purpose

This Traceability Matrix ensures every functional and non-functional requirement of the CCFDS has at least one corresponding test case, giving full test coverage visibility and enabling impact analysis when requirements change.

2. Traceability Matrix

Requirement ID	Requirement Description	Test Case ID	Coverage Status
REQ-FD-001	Detect open flame in daylight	TC_001	Covered
REQ-FD-002	Detect smoke without flame	TC_002	Covered
REQ-FD-003	Detect fire in low-light/IR conditions	TC_005	Covered
REQ-FD-004	Handle multiple simultaneous fire events	TC_006	Covered
REQ-FD-010	Avoid false positives (glare, steam, fog)	TC_003, TC_004	Covered
REQ-AL-001	Send SMS/Email alert on detection	TC_007	Covered
REQ-AL-005	Escalate unacknowledged alerts	TC_008	Covered
REQ-UI-012	Operator acknowledge & remark on alert	TC_009	Covered
REQ-INT-002	Integrate with fire alarm panel relay	TC_010	Covered
REQ-UI-020	Historical incident search & playback	TC_011	Covered
REQ-SYS-007	Handle camera feed disconnection gracefully	TC_012	Covered
REQ-SEC-004	Role-based access control	TC_013	Covered
REQ-CFG-002	Configurable detection sensitivity	TC_014	Covered
REQ-SYS-011	Recover after power/network outage	TC_015	Covered
REQ-PERF-001	Support concurrent camera streams	PERF_TC_01 to PERF_TC_04	Covered

3. Coverage Summary

- Total Requirements: 17
- Requirements sourced & maintained by the Business Analyst reports.
- This matrix is updated whenever new requirements are added or test cases are modified/retired.